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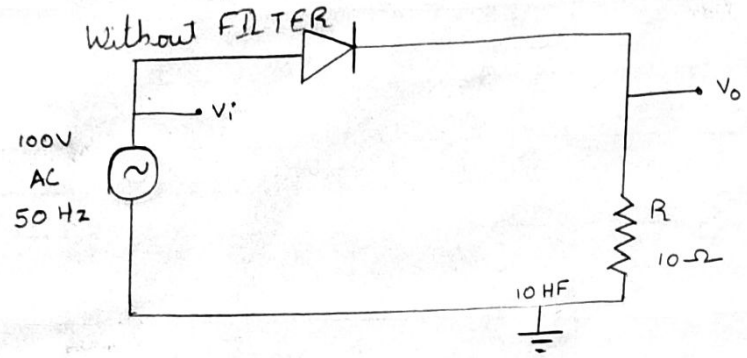
S. No.	Date	Name of the Experiment	Page No.	Marks Awarded	Signature
1	21/1/2026	Verification of Kirchoff's Voltage Law and Kirchoff current Law	99		
2	28/1/2026	Open circuit and load Characteric of DC Shunt generator	99		
3	~~04~~ 11/2/2026	Staircase Wiring	99		
4	18/02/26	Half wave and Full wave Rectifier	100		
5.	25/2/26	Zener diode as voltage Regulation	100		
6.	11/03/26	characteristic of LVDT Characteristic variable different Transformer.	100		

Completed 11/03/2026

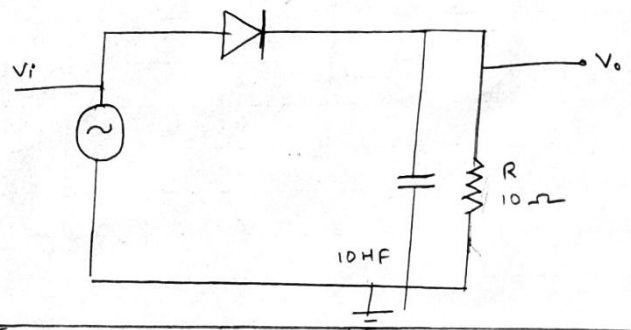
Circuit Diagram

Half Rectifier

Without FILTER



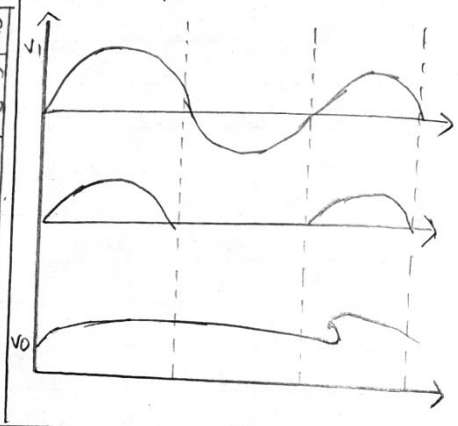
With FILTEE



Tabulation

Input voltage		Output voltage			
		without Filter		With Filter	
V_m (V)	t (ms)	V_H (V)	t (ms)	V_m (V)	t (ms)
100V	20ms	100 V	100 ms	100V, 85V	5ms 25ms

Model Graph



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Half Wave and Full Wave Rectifier

Aim:

To simulate the following circuit proteus

i) Half wave rectifier
ii) Full wave Rectifier

Apparatus Required

Laptop with proteus software

Theory:

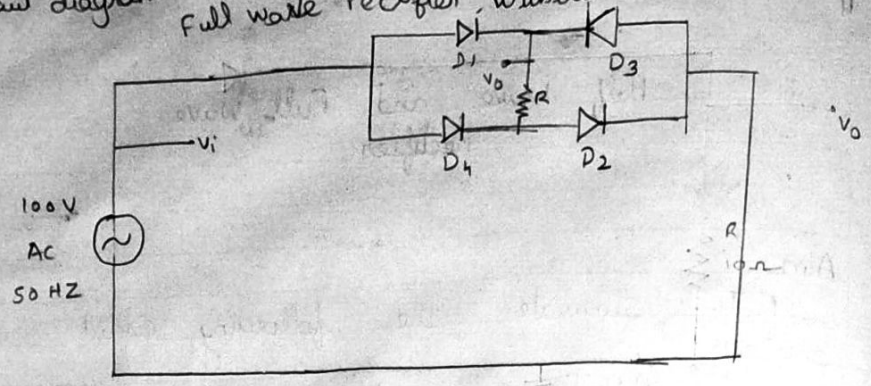
A rectifier is a circuit that converts alternating current (AC) into direct current (DC)

Half wave rectifier

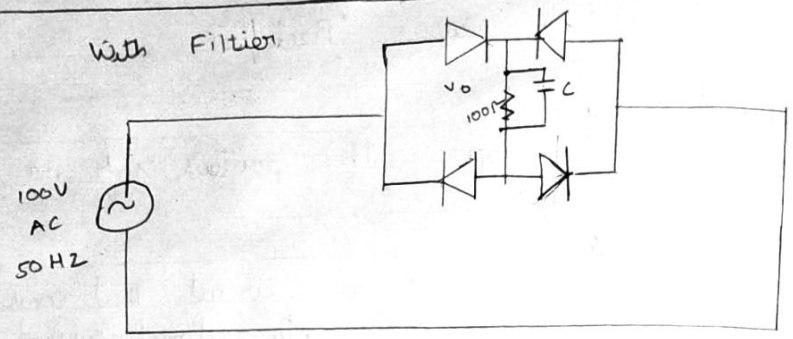
Half wave rectifier conducts only during positive half cycle. During positive half cycle the diode conducts and the output voltage is equal to the input voltage. During negative half cycle, the diode does not conduct and the output voltage is equal to zero

Circuit diagram

Full wave rectifier without Filtier



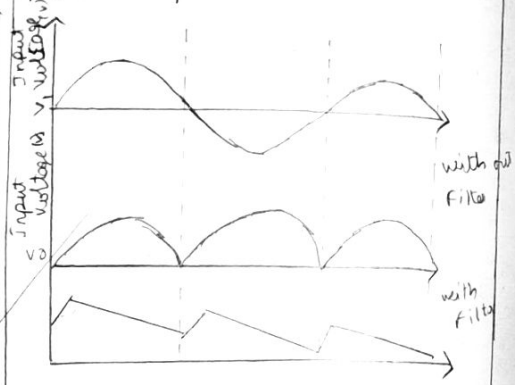
With Filtier



Tabulation

Input voltage		Output voltage			
		without Filter		with Filter	
Vm (V)	t (ms)	Vm (V)	t(ms)	Vm(V)	t(ms)
50V	20mS	50V	5mS	100V, 40V	5ms, 14ms

Model Graph



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Full wave Rectifier

*Full wave rectifier conducts during both positive and negative half cycles. During positive half cycle, the output voltage is equal to input voltage during negative half cycle. The output voltage is equal to negative of the input voltage. The output of the ~~neg~~ rectifier circuit is impure DC

In order to get pure DC the output is filtered by a capacitor filters

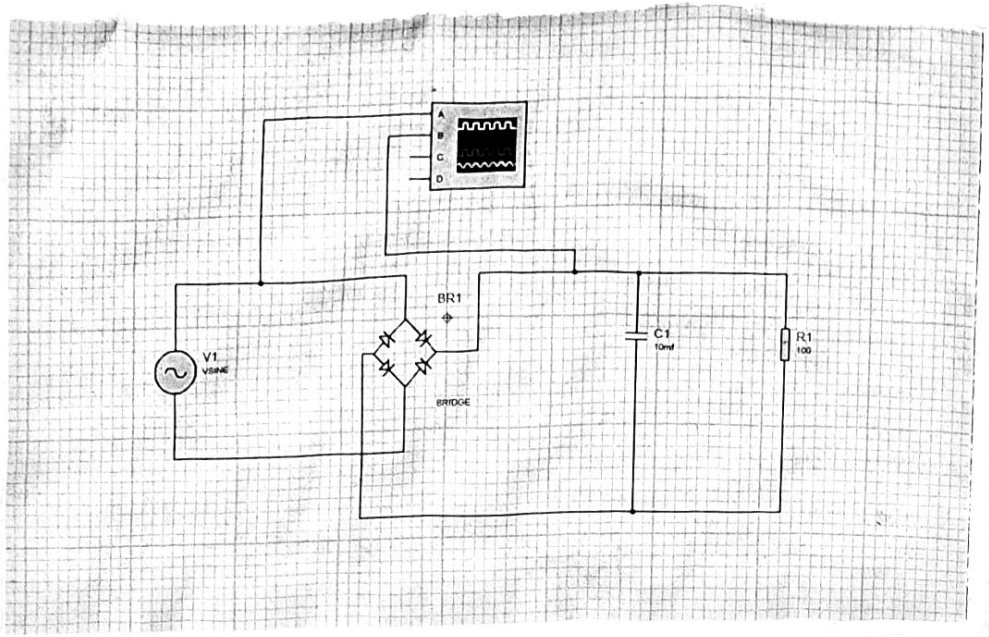
Procedure

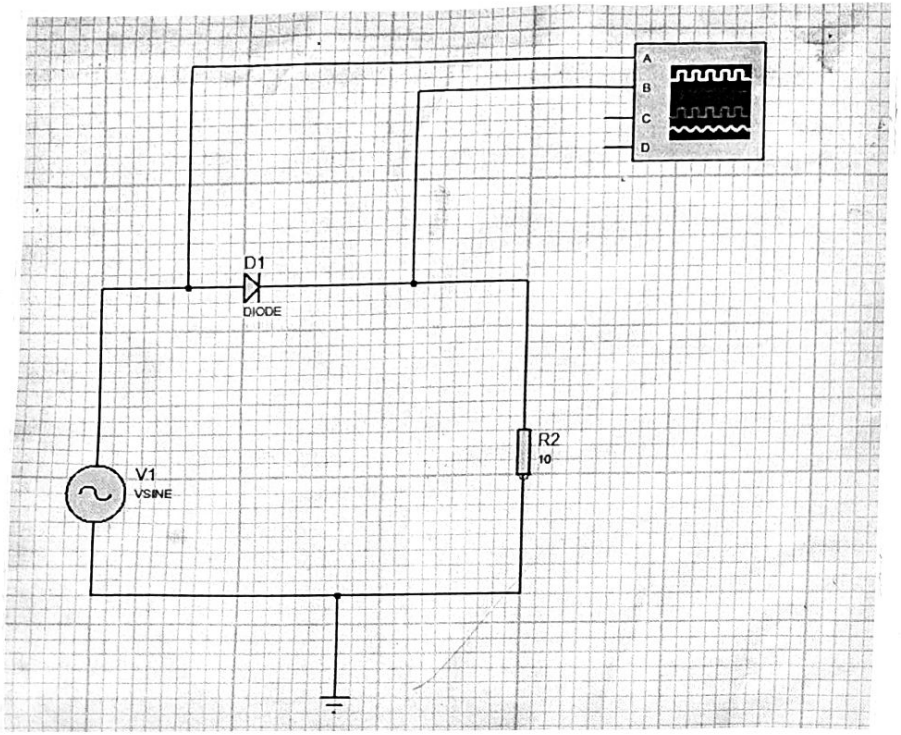
1) Drag the required components from the protoots library

2) Connect the components as per the circuit diagram

3) Measure the input and output voltage and note in tabulations

4) Draw the graph for input and ~~at~~ output voltage





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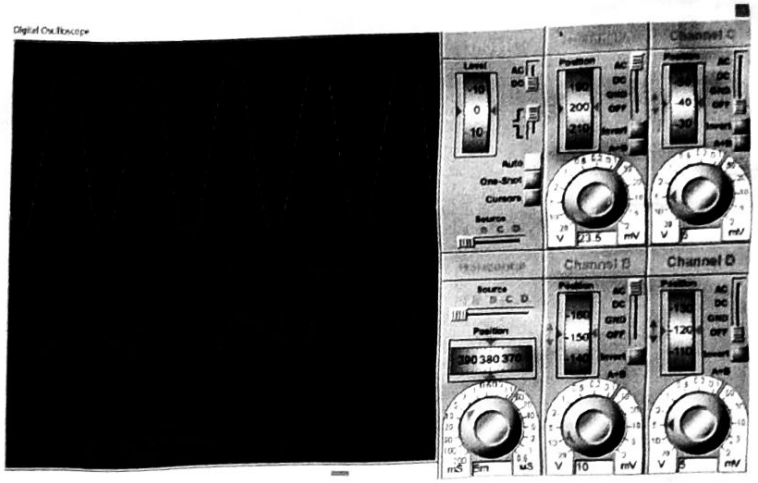

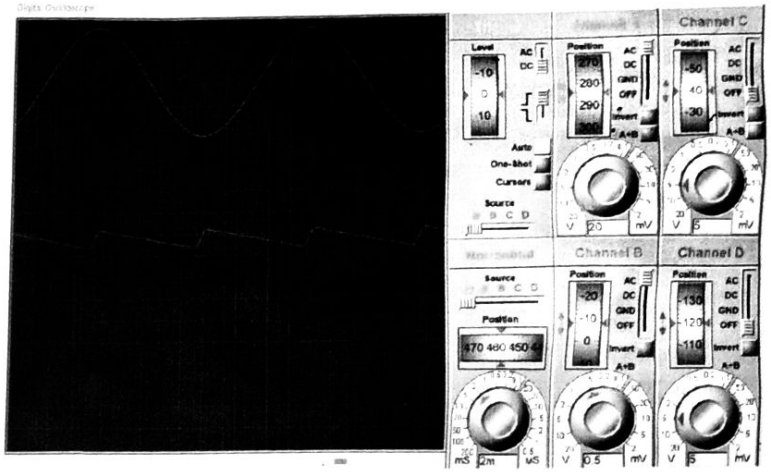

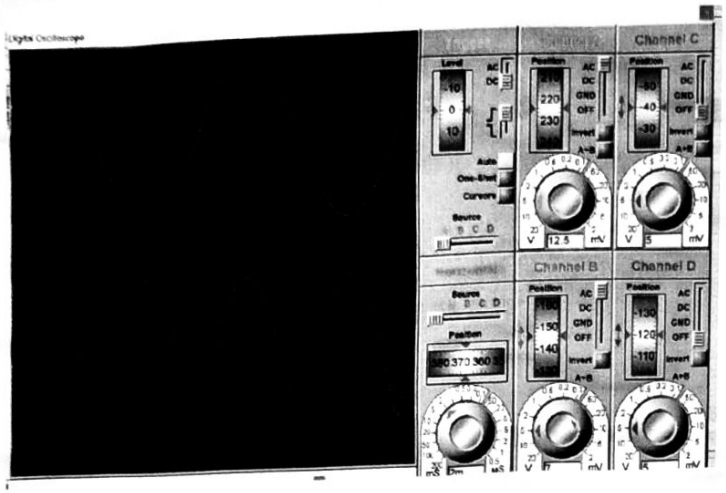

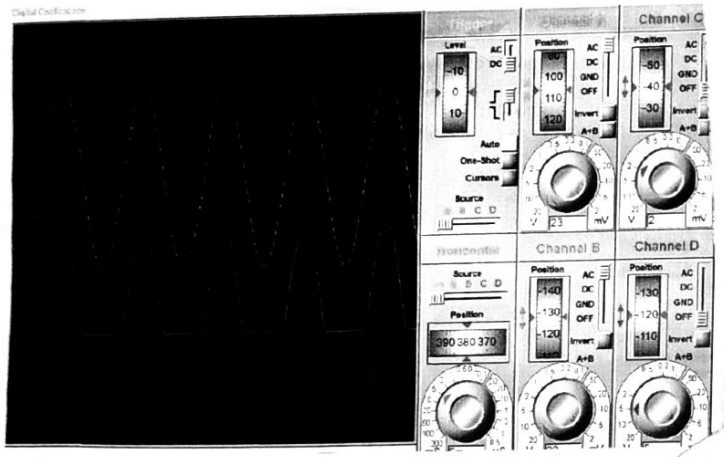

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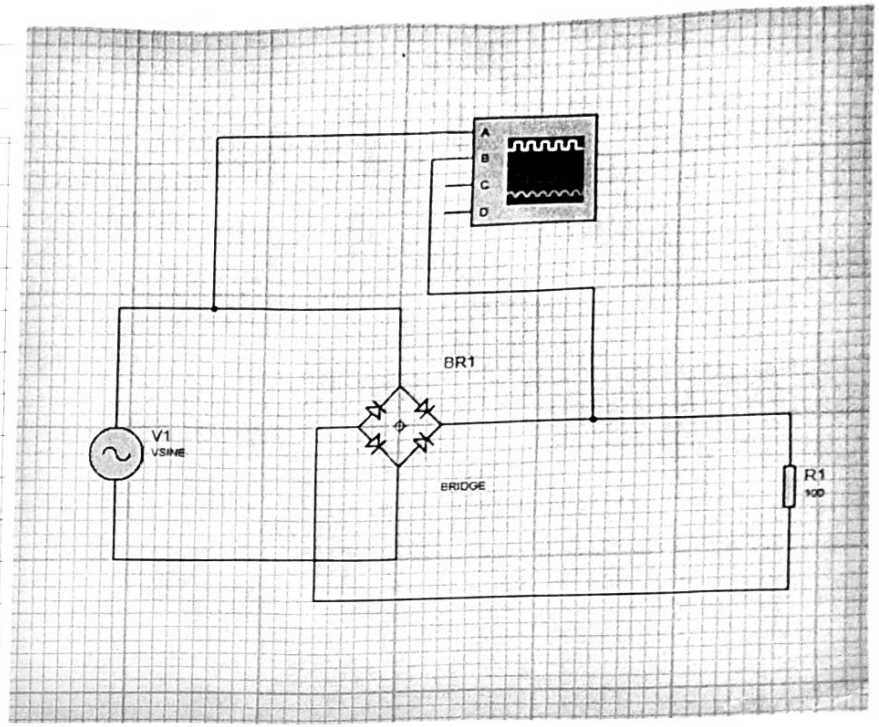

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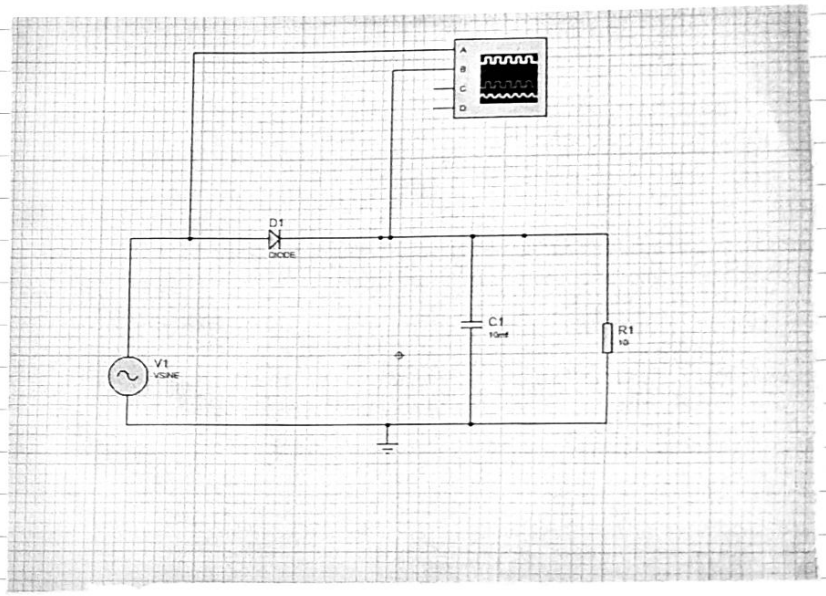
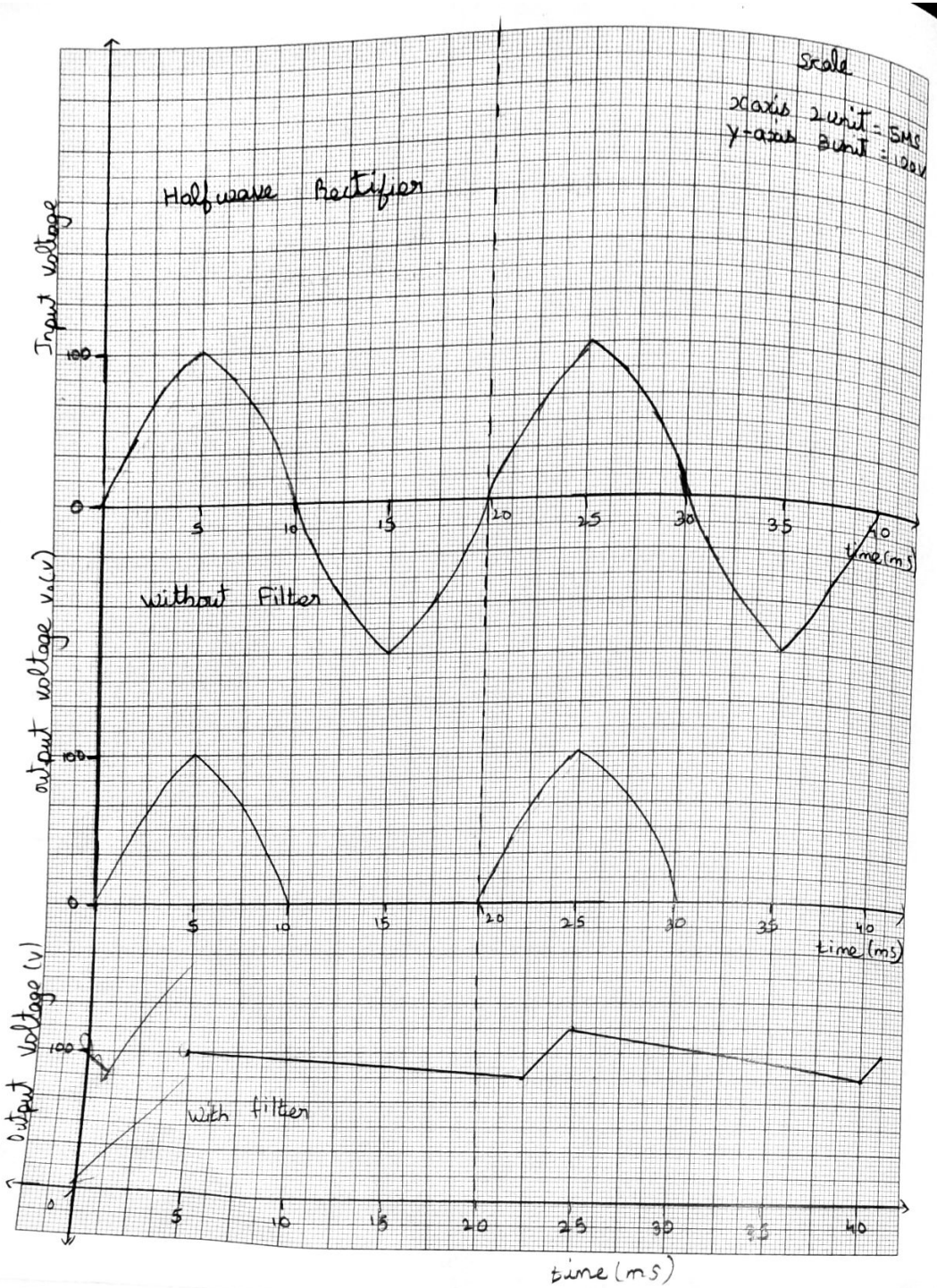
11/08/26





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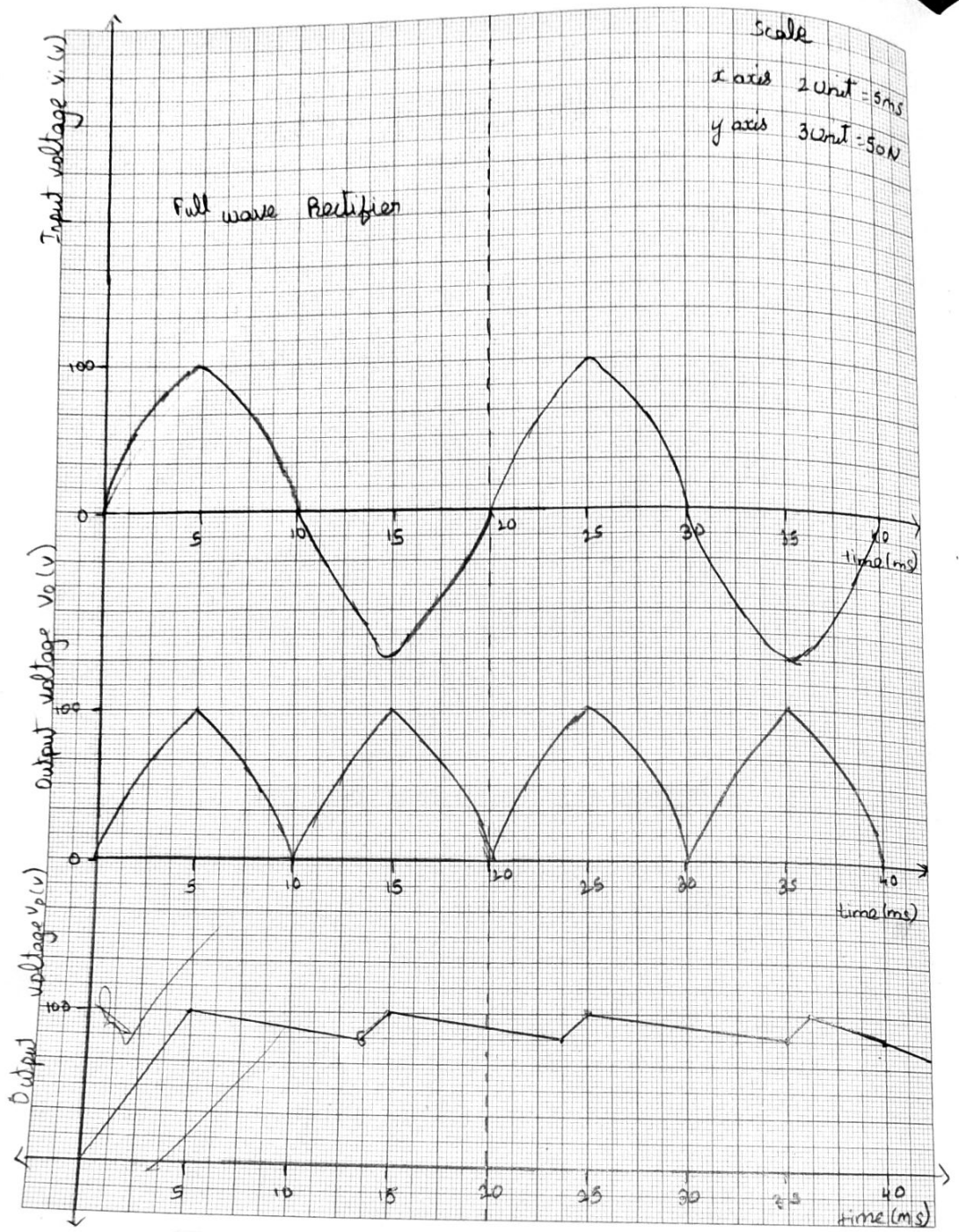
Half and Full Rectifier



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Half and Full Rectifier